

# Advanced Numerical Analysis

## Finite Differences in Infinite-Dimensional & Riemannian Spaces

A Graduate-Level Problem Set | Inspired by M3316 — Dr. Elie Chahine

*"The same river, flowing into a deeper ocean."*

All methods here — divided differences, Richardson extrapolation,  $\theta$ -schemes, CFL stability — are the direct abstract descendants of the finite-dimensional techniques in Chapter 1. The difference is that every scalar becomes an operator, every mesh point becomes a Fourier mode, and every flat space becomes a curved manifold.\*

### Background & Motivation

The classical finite-difference method discretizes a function  $f : [a, b] \rightarrow \mathbb{R}$  over a regular mesh of step size  $h$ , approximates derivatives via divided differences, and solves PDEs by replacing differential operators with algebraic ones. This problem set lifts the entire framework into three progressively harder settings:

- Infinite-dimensional Fourier-Hilbert space**  $\ell^2(\mathbb{Z}, \mathbb{C})$ , where divided differences become bounded operators and Richardson extrapolation operates on operator sequences.
- Complex-conductivity heat equations** in Fourier space, where the  $\theta$ -scheme, Crank-Nicolson, and CFL conditions must be re-derived using functional-analytic tools (Lax-Milgram, Grönwall).
- Riemannian manifolds**  $(\mathcal{M}, g)$ , where the notion of "step  $h$ " becomes a geodesic arc-length, derivatives become covariant derivatives, and curvature introduces a genuinely new source of numerical error with no flat-space analog.

The final exercise combines all three into a single, fully concrete numerical experiment.

### Notational Conventions

| Symbol                                         | Meaning                                                       |     |               |             |                           |
|------------------------------------------------|---------------------------------------------------------------|-----|---------------|-------------|---------------------------|
| $\mathcal{B} = \ell^2(\mathbb{Z}, \mathbb{C})$ | Hilbert space of square-summable complex bilateral sequences  |     |               |             |                           |
| $\hat{f} = (\hat{f}_k)_{k \in \mathbb{Z}}$     | Fourier coefficients of $f \in L^2(\mathbb{T})$               |     |               |             |                           |
| $S_h$                                          | Fourier shift operator: $(S_h \hat{f})_k = e^{ikh} \hat{f}_k$ |     |               |             |                           |
| $D$                                            | Spectral derivative: $(D \hat{f})_k = ik \hat{f}_k$           |     |               |             |                           |
| $\mathcal{B}_s$                                | Sobolev-type subspace: $\{ \hat{f} \in \mathcal{B} : \sum_k$  | $k$ | $\wedge^{2s}$ | $\hat{f}_k$ | $\wedge^{2s} < \infty \}$ |

| Symbol             | Meaning                                                           |  |
|--------------------|-------------------------------------------------------------------|--|
| $(\mathcal{M}, g)$ | Complete Riemannian manifold with Levi-Civita connection $\nabla$ |  |
| $\exp_p, \log_p$   | Riemannian exponential and logarithm at $p \in \mathcal{M}$       |  |
| $K_{\max}$         | Bound on sectional curvature of $\mathcal{M}$                     |  |

## Part I — Fréchet Derivatives in Fourier–Banach Space

Preamble. Let  $\mathcal{B} = \ell^2(\mathbb{Z}, \mathbb{C})$  be the Hilbert space of square-summable bilateral complex sequences  $\hat{f} = (\hat{f}_k)_{k \in \mathbb{Z}}$ , with norm

$$\|\hat{f}\|_{\ell^2} = \left( \sum_{k \in \mathbb{Z}} |\hat{f}_k|^2 \right)^{1/2}.$$

Identify  $\hat{f}$  with the  $L^2(\mathbb{T})$  function via the inverse Fourier transform  $f(x) = \sum_{k \in \mathbb{Z}} \hat{f}_k e^{ikx}$ .

Define the Fourier shift operator  $S_h : \mathcal{B} \rightarrow \mathcal{B}$  by

$$(S_h \hat{f})_k = e^{ikh} \hat{f}_k, \quad h \in \mathbb{R},$$

which corresponds to the spatial translation  $(S_h f)(x) = f(x + h)$  in physical space.

### Exercise 1 — Fourier Forward Difference as a Bounded Operator

Define the forward difference operator in Fourier space:

$$\Delta_h^{\mathcal{F}} : \mathcal{B} \rightarrow \mathcal{B}, \quad \Delta_h^{\mathcal{F}} \hat{f} = S_h \hat{f} - \hat{f}, \quad \text{i.e.,} \quad (\Delta_h^{\mathcal{F}} \hat{f})_k = (e^{ikh} - 1) \hat{f}_k.$$

#### Part (a) — Boundedness and Operator Norm

Prove that  $\Delta_h^{\mathcal{F}} \in \mathcal{L}(\mathcal{B}, \mathcal{B})$  and compute its operator norm:

$$\|\Delta_h^{\mathcal{F}}\|_{\mathcal{L}(\mathcal{B})} = \sup_{k \in \mathbb{Z}} |e^{ikh} - 1|.$$

Determine the value of this supremum explicitly and comment on its behavior as  $h \rightarrow 0$ .

#### Part (b) — First-Order Error in Operator Norm

Define the Fourier derivative approximation operator  $\Phi_h^{\mathcal{F}} = \frac{1}{h} \Delta_h^{\mathcal{F}}$  and the spectral derivative  $D \in \mathcal{L}(\mathcal{B}_1, \mathcal{B})$  by  $(D\hat{f})_k = ik\hat{f}_k$ , which corresponds to  $\partial_x f$  in physical space.

Assuming  $\hat{f} \in \mathcal{B}_1 = \{\hat{f} \in \mathcal{B} : \sum_k k^2 |\hat{f}_k|^2 < \infty\}$ , prove the error bound:

$$\|(\Phi_h^{\mathcal{F}} - D)\hat{f}\|_{\ell^2} \leq \frac{h}{2} \|D^2 \hat{f}\|_{\ell^2},$$

ie., the forward-difference operator is first-order accurate on  $\mathcal{B}_1$ .

**Hint:** Taylor-expand the multiplier  $\frac{e^{ikh}-1}{h} - ik = \frac{(ikh)^2}{2h} e^{i\theta kh}$  for some  $\theta \in (0, 1)$ , component-wise, then apply Plancherel's theorem.

### Part (c) — Second-Order Centered Scheme in $\ell^2$

Define the centered Fourier difference operator:

$$\psi_h^{\mathcal{F}} = \frac{S_h - S_{-h}}{2h}, \quad (\psi_h^{\mathcal{F}} \hat{f})_k = \frac{e^{ikh} - e^{-ikh}}{2h} \hat{f}_k = \frac{\sin(kh)}{h} \hat{f}_k.$$

Prove that for  $\hat{f} \in \mathcal{B}_2$ :

$$\|(\psi_h^{\mathcal{F}} - D)\hat{f}\|_{\ell^2} \leq \frac{h^2}{6} \|D^3 \hat{f}\|_{\ell^2}.$$

Compare this with the classical centered approximation (1.17) of the course, and identify which step of the proof is genuinely new in the infinite-dimensional setting.

## Exercise 2 — Richardson Extrapolation in Operator Space

### Part (a) — First-Order Richardson Lift

Using the spectral representation, show that the explicit Richardson extrapolation of first order extends to operators on  $\mathcal{B}$ :

$$\Phi_h^{1,\mathcal{F}} = 2\Phi_h^{\mathcal{F}} - \Phi_{2h}^{\mathcal{F}},$$

and prove that  $\|(\Phi_h^{1,\mathcal{F}} - D)\hat{f}\|_{\ell^2} = O(h^2)$  for  $\hat{f} \in \mathcal{B}_2$ . Identify precisely the Fourier multiplier of  $\Phi_h^{1,\mathcal{F}}$  and verify the cancellation of the first-order error term.

### Part (b) — $k$ -th Order Operator Recursion

Generalize the Richardson recursion to define the  $k$ -th order extrapolation operator:

$$\Phi_h^{k,\mathcal{F}} = \frac{2^k \Phi_h^{k-1,\mathcal{F}} - \Phi_{2h}^{k-1,\mathcal{F}}}{2^k - 1}.$$

Prove that for  $\hat{f} \in \mathcal{B}_{k+1}$ :

$$\|(\Phi_h^{k,\mathcal{F}} - D)\hat{f}\|_{\ell^2} = O(h^{k+1}).$$

Write the Fourier multiplier of  $\Phi_h^{k,\mathcal{F}}$  explicitly as a polynomial in  $e^{ikh}$  and show it is a  $(k+1)$ -th order approximation of  $ik$  near  $k = 0$ .

### Part (c) — The Spectral Obstruction at Large $|k|$

Show that as  $k \rightarrow \infty$  at fixed  $h > 0$ , the operators  $\Phi_h^{k, \mathcal{F}}$  do not converge to  $D$  in the operator norm  $\|\cdot\|_{\mathcal{L}(\mathcal{B})}$ . Identify precisely the obstruction by comparing the Fourier multiplier  $\frac{e^{ikh}-1}{h}$  against  $ik$  for large  $|k|$ .

**Key observation:** This instability for high-frequency modes is the infinite-dimensional analog of the CFL condition — Richardson extrapolation in infinite dimensions requires a regularization or truncation of high Fourier modes to be stable. This phenomenon has no finite-dimensional counterpart in the classical exercises.

## Part II — Heat Equation in Fourier Space with Complex Conductivity

**Preamble.** Extend the heat equation from Chapter 1, Section 4 to the complex-valued conductivity setting. Consider the PDE on  $\mathbb{T} \times [0, T]$ :

$$\partial_t u - \kappa(x) \partial_{xx} u = f(t, x), \quad u(x, 0) = u_0(x),$$

where  $\kappa \in L^\infty(\mathbb{T}, \mathbb{C})$  satisfies  $\operatorname{Re}(\kappa(x)) \geq \kappa_0 > 0$  almost everywhere. Working in Fourier space with  $u(t, \cdot) = \sum_k \hat{u}_k(t) e^{ikx}$  and  $\kappa(x) = \sum_j \hat{\kappa}_j e^{ijx}$ , the PDE becomes a convolution-operator ODE in  $\ell^2$ .

### Exercise 3 — The $\Theta$ -Scheme in $\ell^2$

The Fourier-space  $\theta$ -scheme reads:

$$\frac{\hat{u}_k^{n+1} - \hat{u}_k^n}{\Delta t} + k^2 \left[ (1 - \theta)(\hat{\kappa} * \hat{u}^n)_k + \theta(\hat{\kappa} * \hat{u}^{n+1})_k \right] = \hat{f}_k^{n+1/2},$$

where  $\hat{\kappa} * \hat{u}$  is Fourier convolution (multiplication by  $\kappa(x)$  in physical space).

### Part (a) — CFL Condition from Spectral Stability

When  $\kappa$  is constant and real, the scheme decouples mode-by-mode. For the explicit scheme ( $\theta = 0$ ):

1. Write the scalar recursion for  $\hat{u}_k^{n+1}$  in terms of  $\hat{u}_k^n$ .
2. Derive the stability condition  $|\hat{u}_k^{n+1}| \leq |\hat{u}_k^n|$  and show it requires  $\kappa k^2 \Delta t \leq 1$  for all  $k$ .
3. Connecting to a spatial mesh of size  $\Delta x = 1/(J+1)$  with highest mode  $k_{\max} \approx \pi/\Delta x$ , recover the classical CFL condition:

$$\kappa \frac{\Delta t}{(\Delta x)^2} \leq \frac{1}{2}.$$

## Part (b) — Implicit Scheme with Complex Conductivity

Let  $\kappa$  be complex-valued with  $\operatorname{Re}(\kappa) \geq \kappa_0 > 0$ . The multiplication operator  $M_\kappa : L^2(\mathbb{T}) \rightarrow L^2(\mathbb{T})$ ,  $(M_\kappa u)(x) = \kappa(x)u(x)$ , is bounded but not self-adjoint. Prove that the implicit scheme ( $\theta = 1$ ) is well-posed for all  $\Delta t > 0$  by:

1. Formulating the implicit step as: find  $u^{n+1} \in H^1(\mathbb{T})$  such that for all  $v \in H^1(\mathbb{T})$ :

$$\langle u^{n+1}, v \rangle_{L^2} + \Delta t \cdot a(u^{n+1}, v) = \langle u^n + \Delta t f^{n+1/2}, v \rangle_{L^2},$$

where  $a(u, v) = \int_{\mathbb{T}} \kappa(x) \partial_x u \overline{\partial_x v} dx$ .

2. Showing  $a(\cdot, \cdot)$  is continuous and coercive on  $H^1(\mathbb{T})$  (using  $\operatorname{Re}(\kappa) \geq \kappa_0$ ), and applying the Lax–Milgram theorem to conclude existence and uniqueness.

## Part (c) — Crank–Nicolson Consistency in Infinite Dimensions

Prove consistency of the Crank–Nicolson scheme ( $\theta = 1/2$ ) in the  $\ell^2$  norm: if  $u \in C^3([0, T], \ell^2(\mathbb{Z})) \cap C^0([0, T], \mathcal{B}_2)$ , show the truncation error satisfies:

$$\|R^n(u)\|_{\ell^2} = O((\Delta t)^2 + (\Delta x)^2),$$

mirroring Proposition 4.7 of the course, but now in infinite dimensions. Identify which classical proof steps carry over verbatim and which require adaptation (e.g., the  $\ell^2$ -norm topology replacing the pointwise  $\|\cdot\|_\infty$  topology).

# Part III — Geodesic Finite Differences on a Riemannian Manifold

**Preamble.** Let  $(\mathcal{M}, g)$  be a complete Riemannian manifold of dimension  $d$ , with Levi-Civita connection  $\nabla$ , Riemannian exponential map  $\exp_p : T_p \mathcal{M} \rightarrow \mathcal{M}$ , and logarithmic map  $\log_p$ . The "step size"  $h > 0$  is now a geodesic arc-length.

## Exercise 4 — Covariant Divided Differences

For a smooth curve  $\gamma : [a, b] \rightarrow \mathcal{M}$  parametrized at equidistant arc-length steps  $h$ , and a smooth function  $f : \mathcal{M} \rightarrow \mathbb{C}$ , define the geodesic forward difference:

$$\Delta_h^g f(\gamma(t)) := f(\exp_{\gamma(t)}(h\dot{\gamma}(t))) - f(\gamma(t)).$$

## Part (a) — Geodesic Taylor Expansion

Prove the Riemannian Taylor expansion: for  $f \in C^3(\mathcal{M})$  and  $v \in T_p \mathcal{M}$ ,

$$f(\exp_p(v)) = f(p) + df_p(v) + \frac{1}{2} \operatorname{Hess}_f(p)(v, v) + O(\|v\|_g^3),$$

where  $\text{Hess}_f = \nabla^2 f$  is the Riemannian Hessian. Use this to show the geodesic forward difference approximates the covariant directional derivative with error:

$$\left| \frac{\Delta_h^g f}{h} - df_p(\dot{\gamma}) \right| \leq \frac{h}{2} \|\text{Hess}_f\|_\infty + O(h^2).$$

### Part (b) — Geodesic Centered Difference and Parallel Transport

Construct a geodesic centered difference operator. The key difficulty:  $\exp_p(hv)$  and  $\exp_p(-hv)$  lie on different tangent fibers, so a naive subtraction of gradient vectors is geometrically meaningless. Define the centered difference using **parallel transport**  $\mathcal{P}_{p \rightarrow q} : T_p \mathcal{M} \rightarrow T_q \mathcal{M}$  along the geodesic:

$$\psi_h^g f(p) := \frac{f(\exp_p(hv)) - f(\exp_p(-hv))}{2h},$$

and show this achieves second-order accuracy  $O(h^2)$ , analogous to equation (1.17) of the course.

### Part (c) — Curvature as a New Source of Error

Show that if the sectional curvature of  $\mathcal{M}$  is bounded  $|K| \leq K_{\max}$ , the curvature contributes an **additional error term** of order:

$$O(h^2 K_{\max} \|df\|_\infty)$$

in the centered difference approximation, which has **no analog in the flat (Euclidean) case**. This term arises from the failure of geodesics to commute: the Jacobi equation

$$\frac{D^2 J}{dt^2} + R(\dot{\gamma}, J)\dot{\gamma} = 0$$

introduces curvature corrections to the "Taylor expansion" of  $f \circ \exp$ .

Conclude: on a flat manifold ( $\mathcal{M} = \mathbb{R}^d, g_{\text{Eucl}}$ ), this term vanishes exactly, recovering the classical result from Chapter 1.

## Part IV — The Grand Boss Problem

### Exercise 5 — Riemannian-Manifold-Driven Schrödinger Heat Flow in $\ell^2(\mathbb{Z}, \mathbb{C})$

This exercise weaves together all previous parts into a single numerical method.

## Setup

Let  $(\mathcal{M}, g)$  be a 2-dimensional Riemannian manifold (the concrete case is the hyperbolic disk  $\mathbb{H}^2$  with constant sectional curvature  $K = -1$ ). Consider a manifold-valued parameter  $p(t) \in \mathcal{M}$  evolving along a prescribed geodesic curve. At each time  $t_n = n\Delta t$ , the anisotropic conductivity operator is defined via the Riemannian metric tensor:

$$\kappa^n(k) = g_{ij}(p(t_n)) k_i k_j \in \mathbb{R}_{>0}, \quad k \in \mathbb{Z}^2,$$

the symbol of the anisotropic Laplacian induced by  $g(p(t_n))$ . The PDE in Fourier space is:

$$\partial_t \hat{u}_k(t) + \kappa^n(k) \hat{u}_k(t) = \hat{f}_k(t), \quad k \in \mathbb{Z}, \quad t \in [t_n, t_{n+1}].$$

## Part (a) — Geodesic Interpolation of the Metric

Between times  $t_n$  and  $t_{n+1}$ , define the geodesically interpolated metric at parameter  $\theta \in [0, 1]$ :

$$g^{n+\theta} = g(\exp_{p^n}(\theta \cdot \log_{p^n}(p^{n+1}))),$$

and the corresponding interpolated conductivity symbol  $\kappa^{n+\theta}(k) = g_{ij}^{n+\theta} k_i k_j$ .

Prove the estimate:

$$\left| \kappa^{n+\theta}(k) - \frac{1}{2}(\kappa^n(k) + \kappa^{n+1}(k)) \right| = O((\Delta t)^2 K_{\max} |k|^2),$$

showing that the midpoint metric approximation introduces a curvature-dependent error scaling with mode number  $|k|^2$ .

## Part (b) — Spectral $\Theta$ -Scheme and Unconditional Stability

Write the fully discrete scheme in  $\ell^2$ :

$$\frac{\hat{u}_k^{n+1} - \hat{u}_k^n}{\Delta t} + (1 - \theta) \kappa^n(k) \hat{u}_k^n + \theta \kappa^{n+1}(k) \hat{u}_k^{n+1} = \hat{f}_k^{n+1/2}.$$

Prove unconditional  $\ell^2$ -stability for the implicit scheme ( $\theta = 1$ ): show there exists a constant  $C > 0$ , independent of  $\Delta t$ ,  $N$ , and the geometry of  $\mathcal{M}$ , such that

$$\|\hat{u}^N\|_{\ell^2} \leq C \left( \|\hat{u}^0\|_{\ell^2} + \Delta t \sum_{n=0}^{N-1} \|\hat{f}^n\|_{\ell^2} \right).$$

Key: Use the positive definiteness of the Riemannian metric:

$$\operatorname{Re}(\kappa^n(k)) = g_{ij}^n k_i k_j \geq \lambda_{\min}(g^n) |k|^2 \geq \kappa_0 |k|^2 > 0.$$

### Part (c) — Curvature-Induced Consistency Error

The truncation error now has two independent sources:

1. **Temporal discretization error:**  $O(|1 - 2\theta| \Delta t)$  from the  $\theta$ -scheme (classical, finite-dimensional source).
2. **Geometric error:** From approximating  $g(p(t))$  by piecewise-geodesic interpolation, contributing  $O(\Delta t \cdot K_{\max} \cdot R^2)$ , where  $R = \sup_n d_{\mathcal{M}}(p^n, p^{n+1})$  is the maximal geodesic displacement per step.

Prove the total consistency error bound:

$$\|R^n(\hat{u})\|_{\ell^2} \leq C(|1 - 2\theta| \Delta t + (\Delta t)^2 + K_{\max} R^2),$$

and conclude: the Crank-Nicolson-geodesic scheme ( $\theta = 1/2$ ,  $R = O(\Delta t)$ ) achieves second-order accuracy if and only if  $K_{\max} = O(1)$ , i.e., the curvature remains bounded along the trajectory of  $p(t)$ .

### Part (d) — Convergence via Infinite-Dimensional Grönwall

Combining parts (b) and (c) via the discrete Grönwall inequality (the infinite-dimensional analog of Theorem 4.3 in the course), prove the convergence theorem:

$$\max_{n=0, \dots, N} \|\hat{u}^n - \hat{u}(t_n)\|_{\ell^2} \leq CT \left( \Delta t + K_{\max} \frac{R^2}{\Delta t} \right).$$

Deduce the optimal geodesic-step-time coupling: setting  $R = O(\Delta t)$  (i.e.,  $p(t)$  moves at bounded Riemannian speed) yields first-order convergence  $O(\Delta t)$ , while the Crank-Nicolson version achieves  $O((\Delta t)^2)$ .

The convergence rate in  $\Delta t$  is therefore limited by the rate at which the manifold parameter  $p(t)$  can be tracked — a genuinely geometric constraint.

### Part (e) — Concrete Numerical Experiment: Hyperbolic Disk

**Geometry.** Take  $\mathcal{M} = \mathbb{H}^2$  (Poincaré disk model,  $|z| < 1$ ) with metric:

$$g_{ij}(z) = \frac{4\delta_{ij}}{(1 - |z|^2)^2}.$$

The sectional curvature is  $K \equiv -1$ , and the geodesic distance from the origin satisfies  $d(0, z) = 2 \operatorname{arctanh}(|z|) \rightarrow \infty$  as  $|z| \rightarrow 1$ .

**Trajectory.** Let  $p(t) = \tanh(t/2) \cdot e^{it}$  (a geodesic spiral approaching the boundary  $\partial\mathbb{H}^2$ ).

**Conductivity symbol.** For mode  $k \in \mathbb{Z}$ , the scalar symbol reduces to:

$$\kappa^n(k) = \frac{4k^2}{(1 - |p^n|^2)^2}.$$

Tasks:

- 1. Implement the Crank–Nicolson scheme for the truncated system  $|k| \leq N = 64$ , with time step  $\Delta t \in \{0.1, 0.05, 0.025, 0.01\}$ .
- 2. For  $f = 0$  and  $u_0(x) = e^{ik_0 x}$  (single-mode initial data), the exact solution is:

$$\hat{u}_{k_0}(T) = \exp\left(-k_0^2 \int_0^T \kappa(p(s)) \, ds\right), \quad \hat{u}_k(T) = 0 \text{ for } k \neq k_0.$$

Compute this integral numerically and compare with the scheme output.

- 3. Verify  $O((\Delta t)^2)$  convergence in  $\ell^2$  for small  $t$  (where  $|p(t)| \ll 1$ , curvature effects are mild).
- 4. Observe degradation: as  $T \rightarrow \infty$  (i.e.,  $|p(T)| \rightarrow 1$ ), the curvature-induced error term  $K_{\max} R^2 / \Delta t$  grows like  $(1 - |p|^2)^{-2}$  — the scheme requires progressively smaller time steps to maintain accuracy near the boundary of the hyperbolic disk. Quantify this numerically.

# Summary: What Makes This Harder Than Chapter 1

The following table contrasts each layer of difficulty.

| Feature                  | Chapter 1 (M3316)                         | This Problem Set                                                                    |   |   |
|--------------------------|-------------------------------------------|-------------------------------------------------------------------------------------|---|---|
| Ambient space            | $\mathbb{R}^n$ , finite mesh              | $\ell^2(\mathbb{Z}, \mathbb{C})$ , infinite-dimensional Hilbert                     |   |   |
| Derivative notion        | Classical $f'(x) \in \mathbb{R}$          | Fréchet derivative / Riemannian covariant derivative                                |   |   |
| Taylor expansion         | Scalar, $\mathbb{R}$ -valued              | Operator-valued; Riemannian geodesic expansion with curvature                       |   |   |
| Richardson extrapolation | Scalar recursion                          | Operator recursion; breaks in operator norm for high (                              | k | ) |
| CFL condition            | $\kappa \Delta t / (\Delta x)^2 \leq 1/2$ | Spectral condition on $\ell^2$ + curvature correction $K_{\max} R^2$                |   |   |
| Consistency error        | $O(\Delta t + (\Delta x)^2)$              | $O(\Delta t + (\Delta x)^2 + K_{\max} R^2)$                                         |   |   |
| Conductivity             | Constant $\kappa \in \mathbb{R}$          | Riemannian metric tensor $g_{ij}(p(t)) \in T^* \mathcal{M} \otimes T^* \mathcal{M}$ |   |   |
| Stability tool           | Discrete maximum principle                | Lax–Milgram theorem + discrete Grönwall in $\ell^2$                                 |   |   |
| New error source         | None beyond truncation                    | Riemannian curvature $K \neq 0$ — absent in all flat-space problems                 |   |   |

The soul of the methods is unchanged. Divided differences, Richardson telescoping,  $\theta$ -schemes, CFL, Lax equivalence — these are the same rivers. The ocean is simply deeper.

*Problem set authored with inspiration from M3316 Numerical Analysis, Dr. Elie Chahine, 2019–2020.*